Marcos Ruiz-Flores Vicente

Backend Software Engineer – APIs, Distributed Systems, AI-Assisted Development
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Summary

Backend Software Engineer at Allianz Technology, building and operating **Java and Spring Boot microservices** in production for an enterprise pricing platform. Outside work I scoped, built and run **Torii**, a live product where I own both the backend APIs and the React front end they serve, from design to rollout and operation. I use **AI-assisted and agentic tools (Claude Code)** every day for implementation, testing and refactoring, with my own project instructions, skills and hooks, and I review and validate everything they generate. I care about reliability, debuggability and the experience of the people using what I build.

Experience

Allianz Technology Nov. 2025 – Present
Software Engineer (Backend) *Barcelona, Spain*

- Build, ship and operate **Java and Spring Boot microservices** for an enterprise insurance pricing platform, owning features from design through rollout and production support.
- Design and integrate **REST APIs** with internal and external systems, and **asynchronous, event-driven** communication between services with **Apache Kafka**, **ActiveMQ** and **Apache Camel**.
- Debug production issues across distributed services, tracing them to the root cause and fixing them before escalating.
- Work with relational and NoSQL data stores through **Spring Data JPA/Hibernate and SQL**, with **Redis** caching on hot paths.
- Ship containerized services with **Docker, Kubernetes and AWS** through **Jenkins CI/CD** pipelines.
- Write unit and integration tests (**JUnit, Mockito**) and take an active part in **code reviews** and technical discussions.
- Use **Claude Code** as part of daily work for implementation, refactoring and tests, critically reviewing all generated code before it is merged.
- Collaborate in English with engineers and business stakeholders in an international **Agile/Scrum** team.

La Salle Campus Barcelona Sep. 2024 – Jul. 2026
Teaching Assistant – Computer Engineering *Barcelona, Spain*

- Supported students in Operating Systems (**C, processes, concurrency**), Systems Administration (**Linux, Bash**) and Databases (**PostgreSQL, Oracle, MongoDB**); reviewed and debugged their code.

Projects

Torii – Flight Deal Search Engine (live at toriitravel.com) | *Java 21, Spring Boot, PostgreSQL, React* 2026

- Scoped, built and operate the whole product on my own: a **Spring Boot API** on Railway serving a **React and TypeScript** UI on Vercel, shipped through **GitHub Actions** CI with automatic deploys on every merge to main.
- Designed the API around what the UI needs: search, price history, saved searches, usage quotas and account endpoints, with consistent **ProblemDetail** errors so the front end can show useful messages.
- Run hundreds of provider calls per search in parallel with **Java 21 virtual threads**, behind one interface with **automatic failover** between external flight APIs.
- Added a **Caffeine cache** per date pair whose TTL depends on how close the trip is, trading data freshness for latency and API cost, plus a stats endpoint to observe hit rates.
- Implemented stateless **Spring Security with JWT**, email verification and password reset (sessions invalidated after a reset), and per-plan quotas returning **HTTP 429**.
- Evolved the schema safely with versioned **Flyway** migrations on **PostgreSQL**, and kept non-critical writes such as price history best-effort so they never break a search.

Education

La Salle Campus Barcelona 2020 – 2026
Bachelor's Degree in Computer Engineering *Barcelona, Spain*

- **Bachelor's thesis:** Impact of Quantum Computing on Cybersecurity. Ran Shor's and Grover's algorithms on IBM Quantum 156-qubit processors (Python, Qiskit) and benchmarked NIST post-quantum algorithms against RSA-2048.

Skills

Languages: Java, Python, TypeScript, JavaScript, C, SQL, Bash
Backend and APIs: Spring Boot, Spring Security, Spring Data JPA, Hibernate, REST, Microservices, FastAPI
Distributed Systems: Apache Kafka, ActiveMQ, Apache Camel, Caching (Redis, Caffeine), Concurrency (Virtual Threads)
Data Stores: PostgreSQL, Oracle, MySQL, MongoDB, Redis, Flyway
Infrastructure and Delivery: Docker, Kubernetes, AWS, Jenkins, GitHub Actions, CI/CD, Linux
Front End: React, TypeScript, Vue.js, Tailwind, HTML, CSS
AI-Assisted Development: Claude Code (project instructions, custom skills, hooks, subagents), reviewing and validating generated code
Testing: JUnit, Mockito, MockMvc, Integration Testing, Code Reviews

Languages

Spanish: Native | **Catalan**: Native | **English**: Professional working proficiency (daily working language)